

6. Op-Amp: Square, Triangular & Sawtooth wave generators:

Advanced Physics Laboratory II (PHY325)

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1 Aim

To set up and study the square, triangular and saw-tooth waveform generators using IC 741.

2 Components Required

The components and equipment required for this experiment are listed in Table 1.

Table 1: List of Required Components and Instrumentation

Component / Instrument	Specification / Value	Quantity
Operational Amplifier	IC 741	2
Resistors	4.7 k Ω	4
	33 k Ω	1
	220 k Ω (Bleeding Resistor)	1
Capacitors	0.47 μ F	2
Variable Resistor	Potentiometer (10 k Ω)	1
Power Supply	Dual Tracking DC Power Supply (± 15 V)	1
Measuring Instruments	Digital Storage Oscilloscope (DSO) with Probes	1
Prototyping	Solderless Breadboard and Connecting Wires	1

3 Theory and Circuit Design

3.1 Square Wave Generator

An op-amp can be configured as a square wave generator (astable multivibrator) using simultaneous positive and negative feedback loops, as shown in Figure 1. A voltage divider made of two equal 4.7 k Ω resistors (R_1 and R_2) at the non-inverting terminal (+) sets the reference trigger voltages ($\pm V_{\text{threshold}}$). An RC timing network at the inverting terminal (-) determines the oscillation time period.

The capacitor C repeatedly charges and discharges toward the op-amp output saturation voltages ($\pm V_{\text{sat}}$). When the voltage across the capacitor crosses the threshold voltage at the non-inverting terminal, the op-amp comparator action forces the output to rapidly switch to the opposite saturation state. This continuous switching produces a steady square wave at the output terminal (Pin 6).

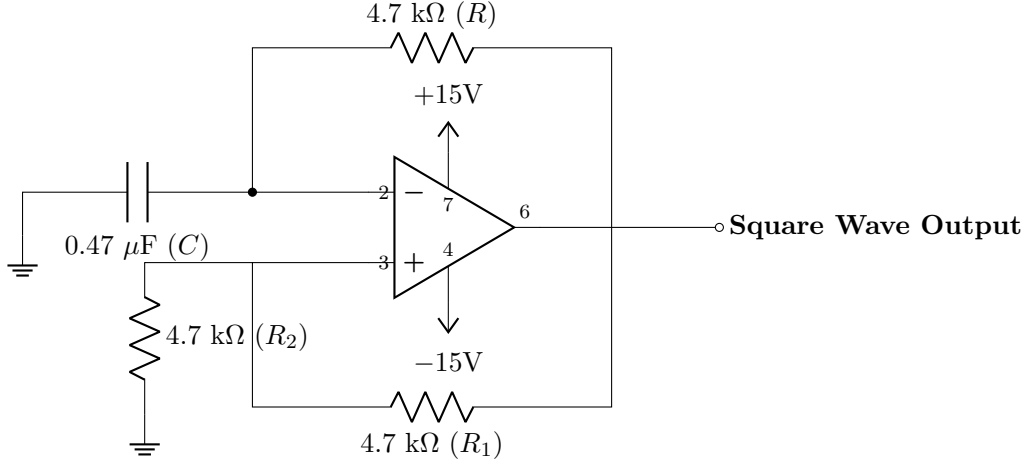


Figure 1: Circuit Diagram 1: Square Wave Generator using an IC 741 op-amp.

For a symmetrical voltage divider where $R_1 = R_2$, the switching thresholds are at $\pm \frac{1}{2}V_{\text{sat}}$, and the theoretical oscillation frequency f is given by:

$$f \approx \frac{1}{2.2 RC} \quad (1)$$

Substituting the experimental component values ($R = 4.7 \text{ k}\Omega$ and $C = 0.47 \text{ }\mu\text{F}$), the theoretical time period T_{theo} and fundamental oscillation frequency f_{theo} are calculated as:

$$T_{\text{theo}} \approx 2.2 RC \approx 2.2 \times (4.7 \times 10^3 \text{ }\Omega) \times (0.47 \times 10^{-6} \text{ F}) \approx 4.86 \text{ ms} \quad (2)$$

$$f_{\text{theo}} = \frac{1}{T_{\text{theo}}} \approx \frac{1}{4.86 \times 10^{-3} \text{ s}} \approx 205.7 \text{ Hz} \quad (3)$$

3.2 Triangle Wave Generator (Integrator)

Integrating a constant voltage over time results in a steadily rising or falling linear ramp. When the square wave produced by Circuit 1 is fed into an op-amp configured as an integrator (Figure 2), the feedback capacitor (C_f) charges and discharges at a constant rate ($\pm I_{\text{in}} = \pm V_{\text{in}}/R_{\text{in}}$). This converts the input square wave into a symmetrical triangular waveform at the output.

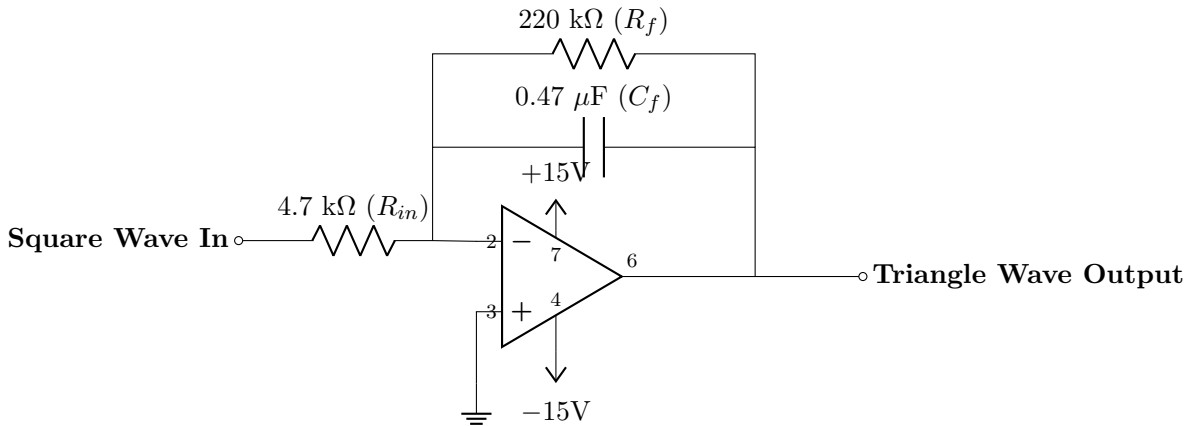


Figure 2: Circuit Diagram 2: Triangle Wave Generator (Integrator with resistor R_f).

In a simple integrator circuit, even a tiny input offset voltage can accumulate across the feedback capacitor over time, eventually driving the op-amp output into saturation (clipping against the power supply rails). To prevent this drift, a large resistor ($R_f = 220 \text{ k}\Omega$) is connected in parallel with the feedback capacitor ($C_f = 0.47 \text{ }\mu\text{F}$). This resistor limits the low-frequency gain and discharges unwanted DC offsets. The output voltage of the integrator is given by:

$$V_{\text{out}}(t) = -\frac{1}{R_{\text{in}}C_f} \int V_{\text{in}}(t) dt \quad (4)$$

Since the integrator operates directly on the incoming square wave, its theoretical time period is identical to that of the input wave:

$$T_{\text{theo, triangle}} = T_{\text{theo, square}} \approx 4.86 \text{ ms} \quad (f_{\text{theo}} \approx 205.7 \text{ Hz}) \quad (5)$$

3.3 Sawtooth Wave Generator

A sawtooth waveform requires completely unequal charging and discharging rates—for instance, a slow linear voltage increase followed by a rapid downward drop. As shown in Figure 3, this asymmetry is achieved by applying an external DC bias voltage (V_{bias}) to the non-inverting terminal (Pin 3) using a potentiometer connected across the $\pm 15\text{V}$ power supply rails.

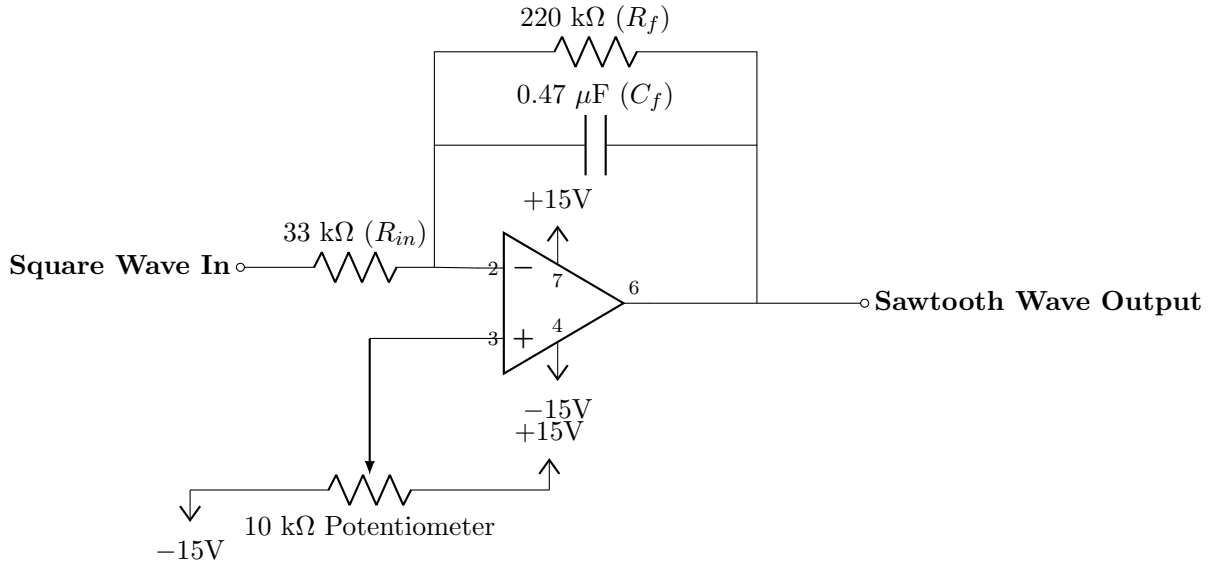


Figure 3: Circuit Diagram 3: Sawtooth Wave Generator with variable DC biasing at Pin 3 and input resistance $R_{\text{in}} = 33 \text{ k}\Omega$.

Applying a DC bias voltage (V_{bias}) to Pin 3 sets the virtual ground at Pin 2 to the same potential ($V_{\text{inverting}} \approx V_{\text{bias}}$). Because of this offset, the current flowing into the capacitor through R_{in} is significantly different when the incoming square wave is at its positive high state ($+V_{\text{sq}}$) versus its negative low state ($-V_{\text{sq}}$):

$$I_{\text{charge}} = \frac{+V_{\text{sq}} - V_{\text{bias}}}{R_{\text{in}}}$$

$$I_{\text{discharge}} = \frac{-V_{\text{sq}} - V_{\text{bias}}}{R_{\text{in}}}$$

When V_{bias} is shifted away from zero, one current magnitude becomes much larger than the other. This causes the capacitor to charge slowly (forming a gradual linear ramp) and discharge quickly (forming a sharp drop), producing a sawtooth shape. Additionally, the input resistor R_{in} is increased from 4.7 k Ω to 33 k Ω to reduce the integration rate so the output voltage does not hit the power supply limits ($\pm 15\text{V}$) and clip.

Even though the charging and discharging rates are unequal, the total time period of each complete cycle is still determined directly by the input square wave:

$$T_{\text{theo, sawtooth}} = T_{\text{theo, square}} \approx 4.86 \text{ ms} \quad (f_{\text{theo}} \approx 205.7 \text{ Hz}) \quad (6)$$

4 Procedure

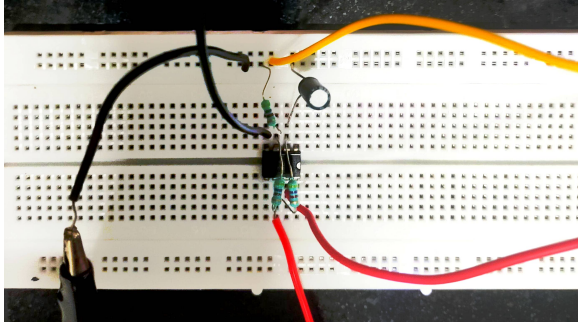
1. **Power Supply Setup:** Connect the dual DC power supply to provide +15V, -15V, and Ground to the power rails of the breadboard.
2. **Square Wave Generator:** Build the circuit around the first IC 741 op-amp as shown in Circuit Diagram 1 (Figure 1). Connect an oscilloscope probe (set to DC coupling) to Pin 6 to observe and verify the generated square wave.
3. **Triangle Wave Generator:** Build the integrator circuit on the second IC 741 op-amp (Circuit Diagram 2, Figure 2). Connect the square wave output from the first circuit to the input of this integrator through a 4.7 k Ω resistor (R_{in}). Include the 220 k Ω resistor in parallel with the capacitor ($C_f = 0.47 \mu\text{F}$) to prevent DC drift. Observe and record the triangular waveform at Pin 6.
4. **Sawtooth Wave Generator:** Modify the integrator circuit by disconnecting Pin 3 from ground and connecting it to the wiper of a 10 k Ω potentiometer placed across the $\pm 15\text{V}$ power rails (Circuit Diagram 3, Figure 3). Increase the input resistor (R_{in}) from 4.7 k Ω to 33 k Ω .
5. **Adjusting the Waveform:** Slowly rotate the potentiometer wiper to change the DC bias at Pin 3. Observe on the oscilloscope how the output waveform shifts from a symmetrical triangle wave into an asymmetric sawtooth wave.

5 Experimental Waveforms

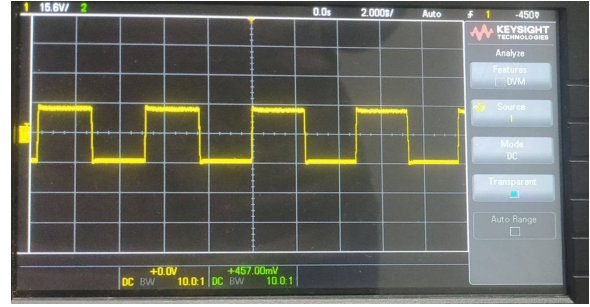
The circuits were built on a breadboard and tested using a Digital Storage Oscilloscope (DSO). For each circuit, photographs of the hardware setup and the output waveform displayed on the oscilloscope were recorded.

5.1 Square Wave Circuit

The first op-amp circuit generated a steady square wave. Figure 4 shows the physical breadboard setup alongside the waveform displayed on the oscilloscope. The output cleanly alternates between positive and negative voltage levels at a frequency close to the theoretical calculation ($\approx 205 \text{ Hz}$).



(a) Breadboard Circuit Implementation

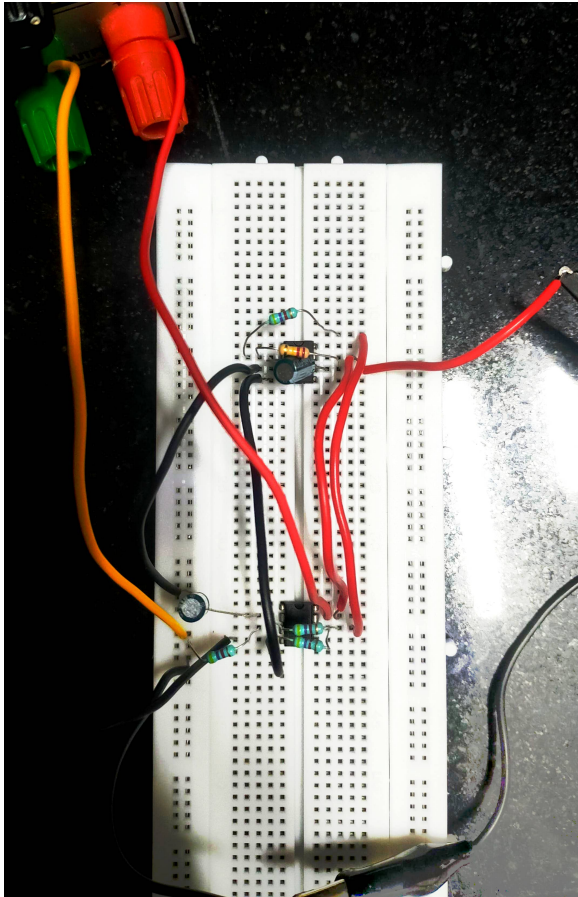


(b) Oscilloscope Signal Capture

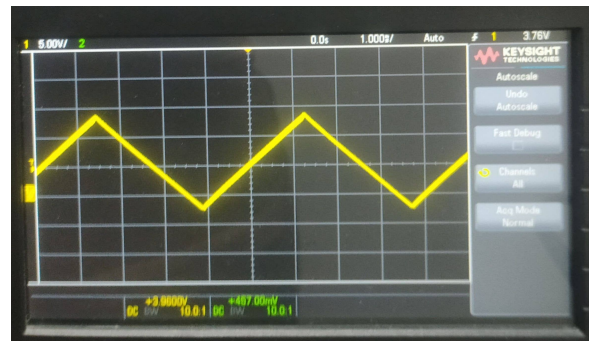
Figure 4: Experimental setup and oscilloscope capture of the square wave generator output at Pin 6.

5.2 Triangular Wave Circuit

Feeding the square wave into the second op-amp integrator produced a clean triangle wave (Figure 5). Since the charging current remains constant during each half of the square wave cycle, the output voltage rises and falls in straight linear slopes. The $220\text{ k}\Omega$ feedback resistor prevented offset voltage drift and kept the signal centered around zero volts without clipping.



(a) Breadboard Circuit Implementation

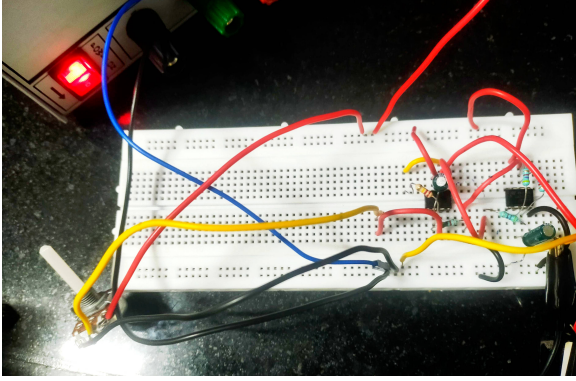


(b) Oscilloscope Signal Capture

Figure 5: Experimental setup and oscilloscope capture of the triangle wave generator output.

5.3 Sawtooth Wave Circuit

By applying an adjustable DC bias voltage at Pin 3 using the potentiometer and changing the input resistor to $33\text{ k}\Omega$, the charging and discharging rates became uneven. As seen in Figure 6, the waveform transformed into a sawtooth wave with a gradual upward slope and a sharp downward drop.



(a) Breadboard Circuit Implementation



(b) Oscilloscope Signal Capture

Figure 6: Experimental setup and oscilloscope capture of the sawtooth wave generator output.

6 Observations and Troubleshooting

6.1 Observed and Theoretical Time Periods

Table 2 compares the theoretical time periods calculated from component values ($T_{\text{theo}} \approx 2.2RC$) against the actual time periods (T_{obs}) and frequencies measured on the oscilloscope.

Table 2: Comparison of Theoretical vs. Observed Time Periods and Frequencies

Waveform	$T_{\text{theoretical}}$	T_{observed}	f_{observed}	Error (%)
Square Wave (Astable)	4.86 ms	4.90 ms	204.1 Hz	0.82%
Triangle Wave (Integrator)	4.86 ms	4.90 ms	204.1 Hz	0.82%
Sawtooth Wave (Biased Int.)	4.86 ms	4.375 ms	228.6 Hz	9.98%

6.2 Circuit Behavior and Troubleshooting

During the setup and testing of the circuits, a few practical behaviors and common measurement issues were observed and resolved:

- **Probing the Input vs. Output:** Connecting the probe to Pin 2 (the inverting input) of the square wave generator instead of Pin 6 shows an exponential charging and discharging voltage instead of a square wave. This occurs because Pin 2 monitors the voltage across the timing capacitor C . To view the true square wave signal, the probe must be connected to the op-amp output on Pin 6.
- **Oscilloscope Coupling Mode:** If the oscilloscope channel is set to AC coupling instead of DC coupling, an internal capacitor is placed in series with the input signal. At around 200 Hz, this acts like a high-pass filter and turns the flat square wave into sharp voltage

spikes (differentiation). The oscilloscope must be set to **DC coupling** to view the correct waveform shape.

- **Incorrect Timebase Setting:** If the oscilloscope timebase is set too slow (such as 2.0 s/div) while viewing a 200 Hz signal, the screen samples too slowly and displays a distorted, stepped “staircase” wave. Adjusting the horizontal timebase to 1–2 ms/div cleanly solves this issue and correctly displays the true wave shape.
- **Op-Amp Saturation (Clipping):** When initially testing the sawtooth circuit using a smaller input resistor ($R_{in} = 4.7\text{ k}\Omega$), too much current entered the integrator. This caused the voltage to rise too fast and hit the $\pm 15\text{V}$ power supply limits, flattening the tops and bottoms of the waveform (clipping). Increasing the input resistor to 33 k Ω slowed down the charging current, keeping the voltage within operational limits and restoring clean sawtooth peaks.

7 Results and Conclusion

In this experiment, three operational amplifier waveform generator circuits were successfully assembled and tested using the IC 741:

1. A stable **square wave** was generated using an op-amp configured as an astable multivibrator, with an observed time period (4.90 ms) closely matching the theoretical calculation (4.86 ms).
2. A clean **triangle wave** was produced by passing the square wave through an integrator circuit. A 220 k Ω feedback resistor proved necessary to prevent offset voltage drift and keep the signal centered about zero volts.
3. An adjustable **sawtooth wave** was created by applying a variable DC bias voltage to Pin 3 of the integrator using a potentiometer and increasing the input resistor to 33 k Ω , demonstrating how voltage biasing effectively controls wave asymmetry.

8 Discussion

8.1 Practical Applications

- **Square Waves:** Widely used as timing clocks in microprocessors, digital logic systems, and pulse-width modulation (PWM) power controls.
- **Triangle Waves:** Commonly employed in audio sound synthesizers, Class-D amplifiers, and linear scanning voltage ramps for test equipment.
- **Sawtooth Waves:** Ideal for horizontal deflection raster scanning in CRT monitors and analog oscilloscopes, where a scanning beam must smoothly trace across a screen and quickly snap back to the starting point.

8.2 Advantages of Op-Amp Oscillators

Using simple operational amplifiers like the IC 741 offers practical laboratory and prototyping benefits:

- **Simplicity & Low Cost:** Generates multiple waveforms with just two inexpensive integrated circuit chips and standard resistors/capacitors.
- **Real-Time Calculus:** Directly demonstrates experimental mathematical integration by converting alternating square pulses into linear triangular slopes.

- **Easy Tuning:** Frequencies and wave asymmetry are easily controlled simply by tweaking resistor values ($T \approx 2.2RC$) or adjusting a simple potentiometer voltage bias.

8.3 Limitations and Modern Alternatives

- **Slew Rate and Analog Drift:** The IC 741 has a modest slew rate ($\sim 0.5 \text{ V}/\mu\text{s}$), causing sharp square wave edges to degrade into sloped trapezoidal lines at higher frequencies above a few kilohertz. Furthermore, simple resistor and capacitor values naturally drift over time with room temperature changes.
- **Modern Improvements:** Today, precision signal generators rarely rely on simple analog feedback. Instead, modern instrumentation uses dedicated function generator chips or **Direct Digital Synthesis (DDS)**, which numerically outputs crystal-accurate waveforms from digital memory using high-speed Digital-to-Analog Converters (DACs).

9 Appendix: Laboratory Logbook Attachment

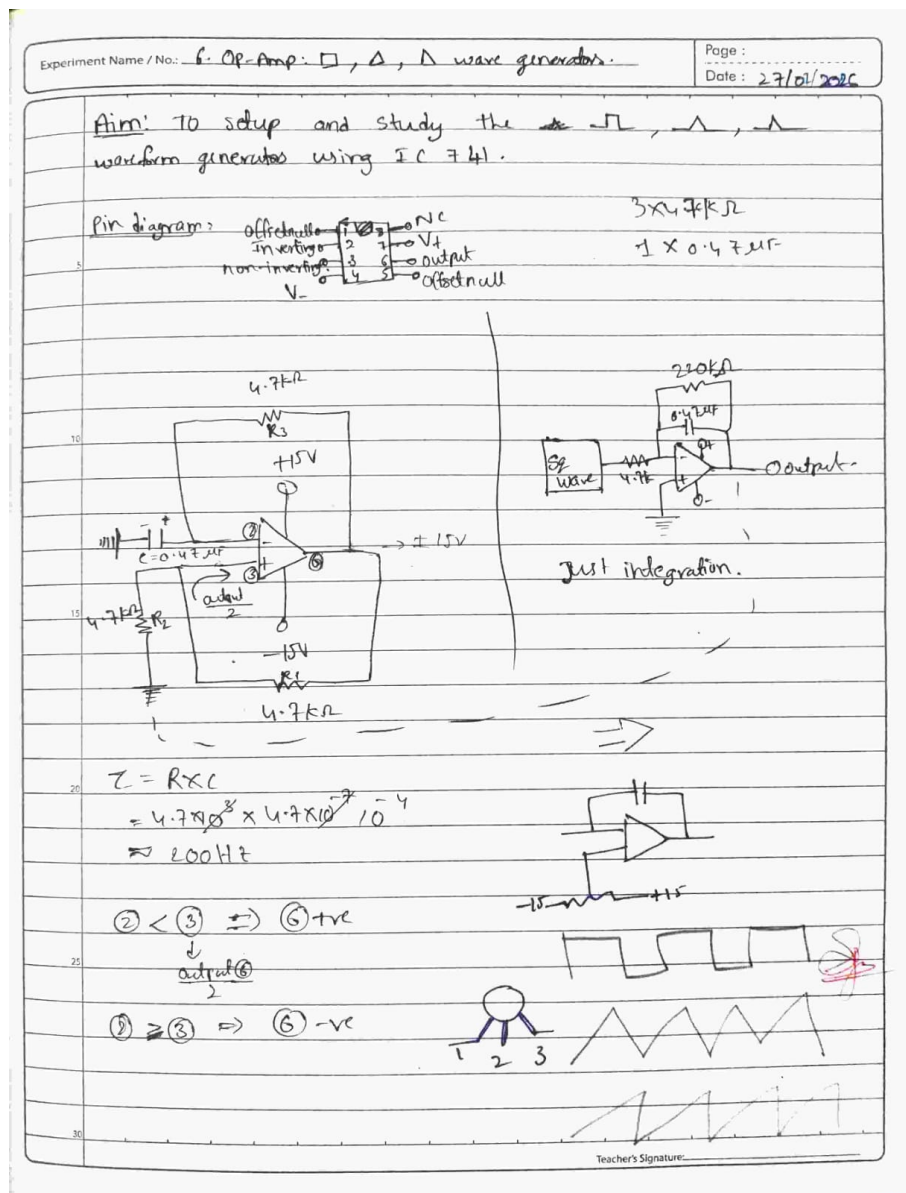


Figure 7: Photograph of the experimental laboratory logbook records.